1. M/J/23/22/Q3

A fixed mass of gas in a glass tube is trapped by a seal at one end of the tube and by a column of mercury. The mercury is free to move within the tube.

The tube is rotated slowly from the vertical as shown in Fig. 3.1 to the horizontal as shown in Fig. 3.2. The volume of the gas increases and its temperature remains constant.

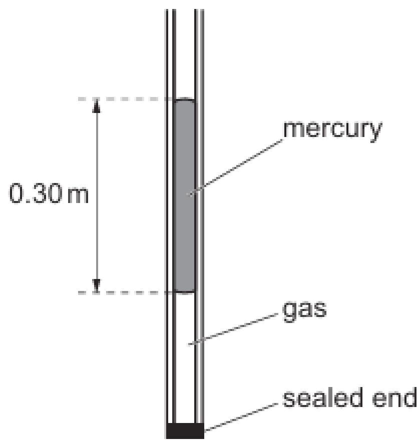

Fig. 3.1 (not to scale)

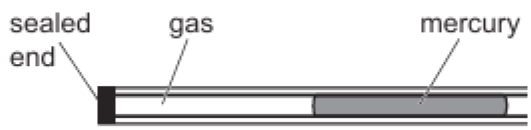

Fig. 3.2 (not to scale)

(a) (i) Describe why rotating the tube changes the pressure of the gas in the sealed end.

....

....

.... [1]

(ii) Explain, using ideas about particles, why the pressure of the gas decreases when its volume increases.

....

....

....

.... [3]

(b) In Fig. 3.1 the length of the mercury column is 0.30 m.

The density of mercury is $14\,000\,kg/m^3$.

Atmospheric pressure is $1.0 \times 10^5\,Pa$.

Calculate the pressure of the gas in the tube.

pressure = ... Pa [3]

(c) The pressure of a different sample of gas changes at constant temperature.

Fig. 3.3 shows one point, marked X, on a graph of pressure against volume for the gas sample.

At X the pressure of the gas is P_0 and its volume is V_0.

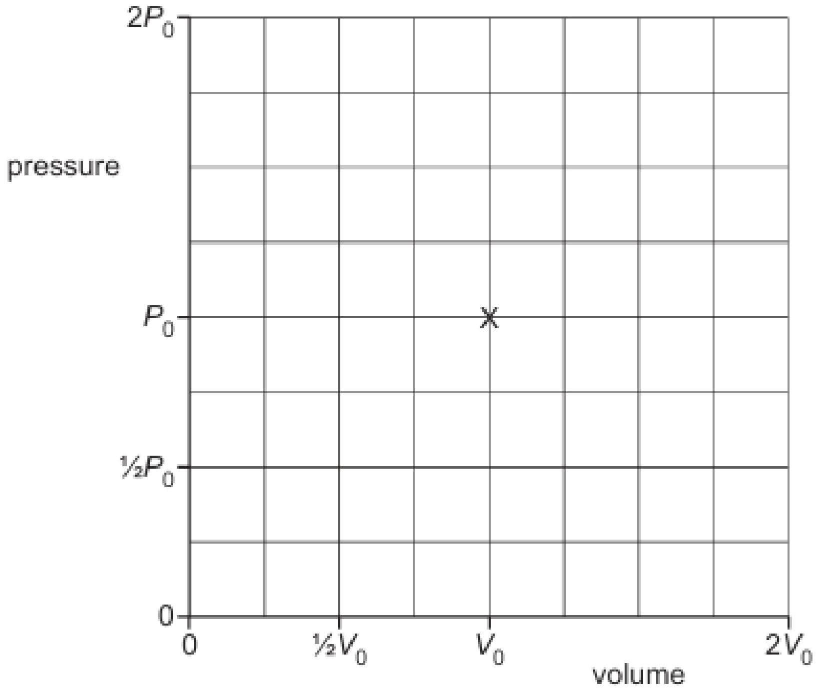

Fig. 3.3

On Fig. 3.3, sketch the graph as the pressure of the gas decreases from $2P_0$ to $\frac{1}{2}P_0$. [2]

[Total: 9]

2. M/J/22/22/Q3

Fig. 3.1 shows a syringe mounted vertically in a block of wood and sealed at one end. A plunger is free to move inside the syringe.

There is trapped air in the syringe.

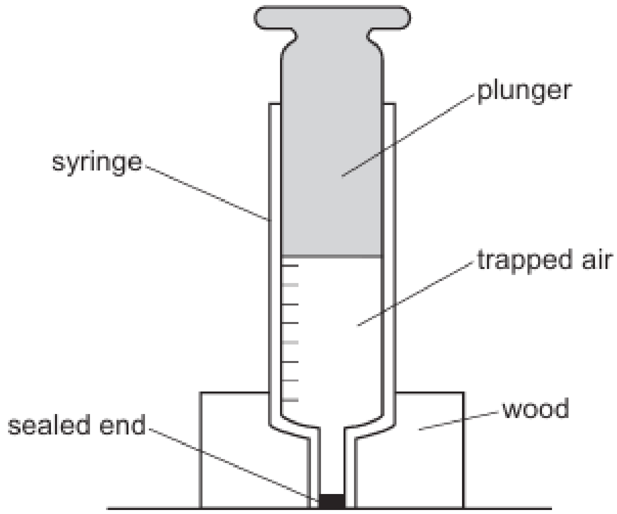

Fig. 3.1

The air inside the syringe exerts a pressure on the walls of the syringe.

(a) Define the term pressure.

....

....

.... [1]

(b) Explain how the air molecules in the cylinder of the syringe create a pressure.

....

....

....

....

.... [3]

(c) A 10 N weight is placed on top of the plunger. The plunger moves down slowly so that the temperature of the air inside the syringe does not change.

Before the weight is placed on top of the plunger:

- the pressure of the air inside the syringe is 1.0×10^5 Pa
- the volume of the air is 50 cm^3.

The cross-sectional area of the plunger is 1.2×10^{-4} m^2.

(i) Calculate the pressure of the air in the syringe after the plunger stops moving.

pressure = ... [2]

(ii) Calculate the volume of air inside the syringe after the plunger stops moving.

volume = ... [2]

[Total: 8]

3. O/N/21/21/Q9(a)

Liquid-in-glass thermometers use the expansion of a liquid to indicate the temperature.

(a) Fig. 9.1 shows the molecular structure of a solid and a gas.

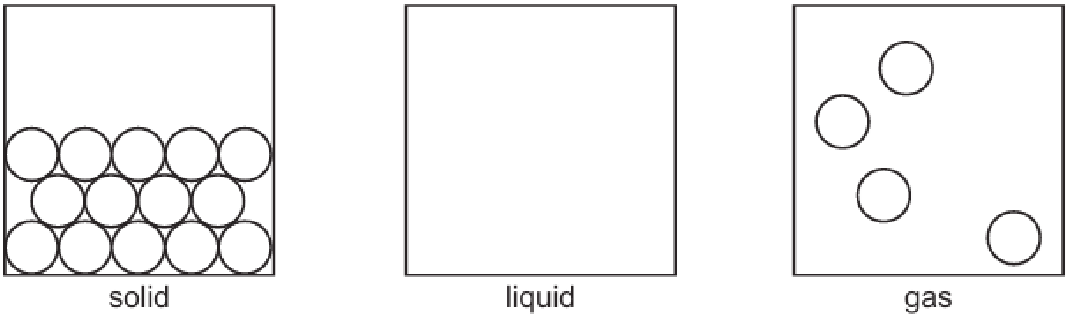

Fig. 9.1

(i) In the middle box of Fig. 9.1, sketch a diagram to show the molecular structure of a liquid. [2]

(ii) Explain why it is easier to compress a gas than to compress a solid.

....

....

.... [2]

4. O/N/20/21/Q8(c)

An underwater depth gauge contains a small cylinder as shown in Fig. 8.1. Gas is trapped inside the cylinder by a piston. The piston is free to move.

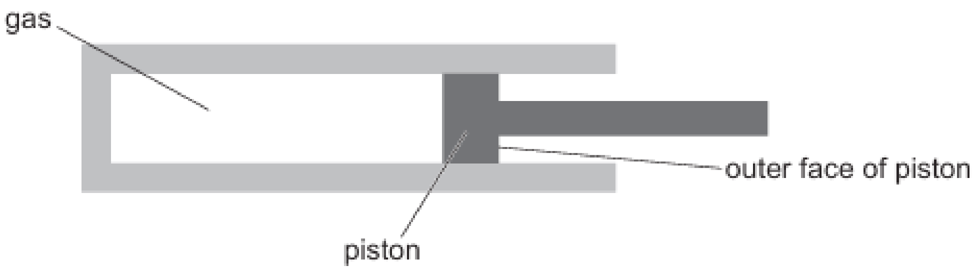

Fig. 8.1

The outer face of the piston is in contact with the water.

As the depth gauge is lowered into the water, the piston moves into the cylinder. This moves a needle on a dial to indicate the depth of the gauge in the water.

(i) Explain why the piston moves into the cylinder.

....

....

....

....

.... [3]

(ii) The temperature of the gas does not change as the piston moves into the cylinder.

Explain, in terms of molecules, what happens to the pressure of the trapped gas as the piston moves into the cylinder.

....

....

....

.... [3]

(iii) At the surface of the water, the volume of the trapped gas in the depth gauge is V_0.

On Fig. 8.2, sketch a graph to show how the volume of trapped gas decreases as the gauge is lowered into the water.

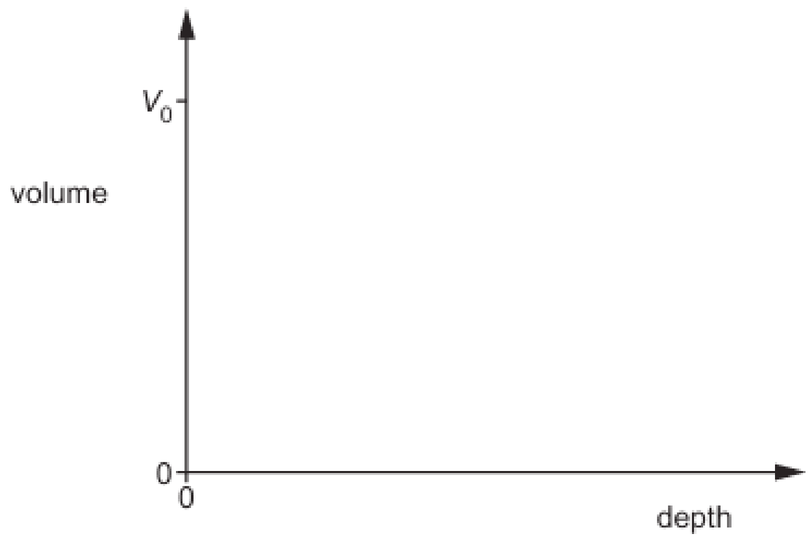

Fig. 8.2

[2]

5. M/J/20/21/Q5

(a) Use the relationship between pressure, force and area to explain why it is harder to cut something with a blunt knife than with a sharp knife.

....

....

.... [2]

(b) Experimental measurements on gas pressures were made by Robert Boyle.

He showed that $p_1V_1 = p_2V_2$ where p_1 and p_2 are the initial and final pressures of a gas, and V_1 and V_2 are the initial and final volumes of the gas.

(i) State **two** quantities that must remain constant when this equation is used.

1. ....

2. ....

[2]

(ii) Fig. 5.1 shows the molecules of a gas as the volume of the gas is halved.

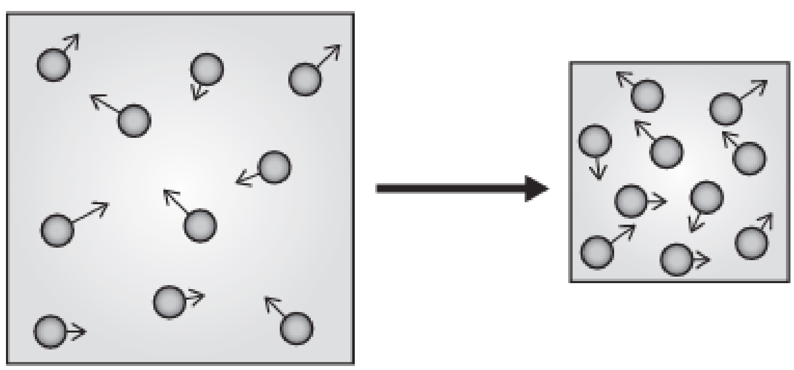

Fig. 5.1 (not to scale)

The equation suggests that when the volume of a gas halves the pressure doubles.

Using ideas about molecules, explain why this happens.

....

....

....

.... [2]

[Total: 6]

6. M/J/19/21/Q4

Fig. 4.1 shows a cylinder with a piston that contains a gas.

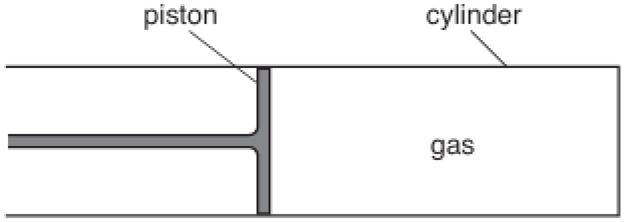

Fig. 4.1

(a) A liquid occupies a much smaller volume than a gas which has the same number of molecules.

Explain why there is this difference.

....

....

.... [2]

(b) The volume of the gas changes from V_1 to V_2 as the piston moves into the cylinder. This increases the pressure from p_1 to p_2. The temperature remains the same.

(i) State the formula that relates V_1, V_2, p_1 and p_2.

.... [1]

(ii) The initial pressure of the gas in the cylinder is 1.2×10^5 Pa and its initial volume is $100\,cm^3$. The cross-sectional area of the piston is $5.0\,cm^2$.

The piston is pushed in a distance of 8.0 cm.

Calculate the final pressure of the gas in the cylinder.

pressure = ... [3]

(iii) Explain, using ideas about molecules, why the pressure of the gas changes.

....

....

.... [1]

[Total: 7]

7. M/J/17/22/Q5

When a sample of a solid is heated, it becomes a liquid and eventually becomes a gas.

(a) Fig. 5.1 shows the arrangement of the molecules in the solid.

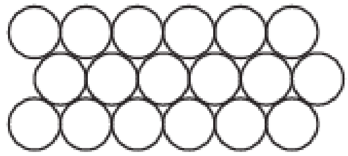

Fig. 5.1

In the space below, draw the arrangement of the molecules in the liquid.

[1]

(b) Complete the table by describing the motion of the molecules in the solid, liquid and gas.

	motion of the molecules
solid	
liquid	
gas	

[3]

(c) Suggest why a gas is able to fill a container but a solid has a fixed shape.

....

....

....

....

....[2]

8. M/J/13/21/Q10

(a) Describe the arrangement and the movement of the molecules

1. in liquid water,

....

....

....

.... [2]

2. in ice.

....

....

....

.... [2]

(b) Ice at 0 °C becomes water at 0 °C.

State what, if anything, happens to the kinetic energy and the potential energy of the molecules as this happens.

kinetic energy: ....

potential energy: ....
[1]

9. M/J/09/Q4

Fig. 4.1 shows the arrangement of molecules in a solid and in a liquid.

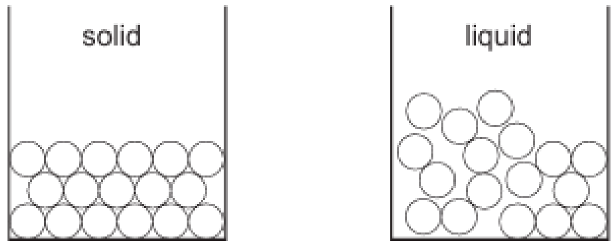

Fig. 4.1

(a) State one difference between the two arrangements.

....

....

.... [1]

(b) By writing about the forces between molecules and the motion of molecules, explain why

(i) the molecules of a solid and of a liquid have different arrangements,

....

.... [1]

(ii) the evaporation of a liquid cools the liquid,

....

....

.... [2]

(iii) the rate of evaporation is greater when a liquid is hotter.

....

....

.... [2]